Scheme of Valuation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCECO300 - ECONOMETRICS -I

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Part A: 1 mark each (10 × 1 = 10)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	3	B	5	B	7	B	9	A
2	D	4	A	6	B	8	D	10	A

Part B: 3 marks each (any 10 × 3 = 30)

11. Panel data

- Definition: the same cross-sectional units (individuals, firms, countries) observed over several time periods
- Uniqueness: it combines cross-section and time-series dimensions, which controls for unobserved heterogeneity, gives more variation, less collinearity and more degrees of freedom, and allows dynamic analysis.

12. Heteroscedasticity

- Variance of the error term is not constant across observations: $\mathrm{Var}(u_i) = \sigma_i^2$
- Example, such as consumption variance rising with income
- Consequence: OLS is unbiased but inefficient, and standard errors are biased

13. Scope of Econometrics

- Definition and nature: economic theory, mathematics and statistics combined
- Any three areas: testing economic theory, estimating parameters/relationships, forecasting, policy evaluation and formulation

14. Negative autocorrelation

- Error terms in successive periods are negatively correlated: $\text{Cov}\,(u_t, uu_{t-1}) < 0,\ \rho < 0$
- A positive error tends to be followed by a negative one, giving a sign-alternating pattern (DW > 2)

15. Multiple regression model

- Model: $C = \beta_0 + \beta_1 Y + \beta_2 W + u$, with C as consumption expenditure, Y as income and W as wealth (or family size)
- Explanation of variables and error term
- Expected signs and interpretation: β_1 is the marginal propensity to consume $(0 < \beta_1 < 1)$, and $\beta_2 > 0$

16. Smaller standard error

- Higher precision and reliability, with less sampling variability
- Narrower confidence intervals and larger t-values
- The estimator is closer to the true parameter, i.e., more efficient

17. Assumptions of t-test

- Population or errors are normally distributed
- Random sampling with independent observations
- Equal variances (two-sample test) or unknown population variance
- Data on an interval or ratio scale

18. Sample Regression Function

- Definition: the sample counterpart of the PRF, giving the estimated relationship from sample data
- Equation: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$
- Explanation of terms, with residual $\hat{u}_i = Y_i - \hat{Y}_i$

19. Econometric methodology

- Meaning: applying statistical and mathematical methods to economic data to give empirical content to theory
- Main steps, any six: statement of theory, specification of the mathematical model, specification of the econometric model, data collection, estimation, hypothesis testing, forecasting, policy use

20. "Independent variables should be truly independent"

- Strictly, the regressors must not be perfectly linearly related (no perfect multicollinearity)
- Some correlation among regressors is acceptable, but high correlation inflates standard errors and makes estimates unstable
- What is essential is that regressors are uncorrelated with the error term (exogeneity). So, the statement is only partly correct

21. OLS estimators

- $\hat{\beta}_1 = \Sigma(X_i - \bar{X})(Y_i - \bar{Y}) / \Sigma(X_i - \bar{X})^2 = \Sigma x_i y_i / \Sigma x_i^2$
- $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

22. Power of a test

- Probability of rejecting a false null hypothesis, equal to $1 - \beta$, where β is the probability of a Type II error
- It increases with larger sample size, larger significance level and larger effect size

23. Meaning of $R^2 = 0.80$

- 80% of the variation in the dependent variable is explained by the regressors
- The remaining 20% is unexplained and is due to other factors or the error term
- Indicates a good fit, but does not imply causation

Part C: 6 marks each (any 5 × 6 = 30)

24. Assumptions of CLRM

1. Regression is linear in parameters
2. X values are fixed in repeated sampling, or independent of the error
3. Zero mean of the error: $E(u_i) = 0$
4. Homoscedasticity: $Var(u_i) = \sigma^2$
5. No autocorrelation: $Cov(u_i, u_j) = 0$
6. Zero covariance between u_i and X_i
7. Number of observations exceeds the number of parameters
8. Variability in the X values
9. Correct model specification
10. No perfect multicollinearity (multiple regression)
11. Normality of the errors (for inference)

25. Dummy variables

- Meaning: binary $(0/1)$ variables that quantify qualitative attributes
- Gender example: $Y_i = \alpha + \beta D_i + u_i$, with $D = 1$ for male and 0 for female. α is the mean for the base category and β is the differential effect
- Religion and marital status
- For m categories use $(m - 1)$ dummies, with one category as the base. Using m dummies causes the dummy variable trap, i.e., perfect multicollinearity
- Interpretation of coefficients relative to the base category
- Mention of interaction (slope) dummies, or a worked example of a multi-category case

26. Four simple regression models

Any four models, with equation, variable definitions and expected sign, at 1.5 each:

- Keynesian consumption function: $C = \beta_0 + \beta_1 Y + u,\ (0 < \beta_1 < 1)$
- Demand function: $Q = \beta_0 + \beta_1 P + u,\ (\beta_1 < 0)$
- Phillips curve: $\pi = \beta_0 + \beta_1 U + u,\ (\beta_1 < 0)$
- Cobb-Douglas production function: $\ln Q = \beta_0 + \beta_1 \ln L + u$
- Other valid models, such as the Engel curve, supply function or money demand, are also acceptable.

27. Multicollinearity, standard errors and t-values

- Meaning of multicollinearity: high correlation among regressors
- Effect on standard errors: $\mathrm{Var}(\hat{\beta}_j) = \sigma^2 / [\Sigma x_j^2 (1 - R_j^2)] = (\sigma^2 / \Sigma x_j^2) \times \mathrm{VIF}$. As R_j^2 approaches 1, variance and standard error become very large, and confidence intervals widen
- Effect on t-values: since $t = \hat{\beta}_j / \mathrm{se}(\hat{\beta}_j)$, inflated standard errors give small, insignificant t-values even when R^2 is high and the F-test is significant. This leads to Type II errors and unstable estimates

28. Sampling distribution, point and interval estimation

- Sampling distribution: the probability distribution of an estimator over repeated samples, e.g., $\hat{\beta}_1 \sim N(\beta_1, \sigma^2 / \Sigma x_i^2)$
- Point estimation: a single sample value $(\hat{\beta}_0, \hat{\beta}_1)$ used to estimate the parameter
- Interval estimation: $\hat{\beta} \pm t(\alpha/2, n - 2) \cdot \mathrm{se}(\hat{\beta})$, a range with a stated confidence level
- Role of the standard error: $\mathrm{se}(\hat{\beta}_1) = \hat{\sigma} / \sqrt{\Sigma x_i^2}$ measures precision, determines the width of the interval and is used in t-tests

29. Importance of the critical region

- Definition: the set of test-statistic values for which H_0 is rejected
- Its size equals the significance level α, the probability of a Type I error
- Decision rule: reject H_0 if the statistic falls in the critical region, otherwise do not reject

- One-tailed versus two-tailed regions, with critical values
- Importance: it gives an objective rule, controls the Type I error rate and affects the power of the test
- A diagram of the normal or t distribution earns the marks for the relevant point.

30. R^2 using ESS and TSS

- Definitions: $\mathrm{TSS} = \Sigma(Y_i - \bar{Y})^2, \mathrm{ESS} = \Sigma(\hat{Y}_i - \bar{Y})^2, \mathrm{RSS} = \Sigma \hat{u}_i{}^2$
- Decomposition: $\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}$
- Formula: $R^2 = \mathrm{ESS}/\mathrm{TSS} = 1 - \mathrm{RSS}/\mathrm{TSS}$
- Interpretation: lies between 0 and 1, and measures the proportion of variation in Y explained by the model. A higher value means a better fit